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Software Engineering
Lab-1

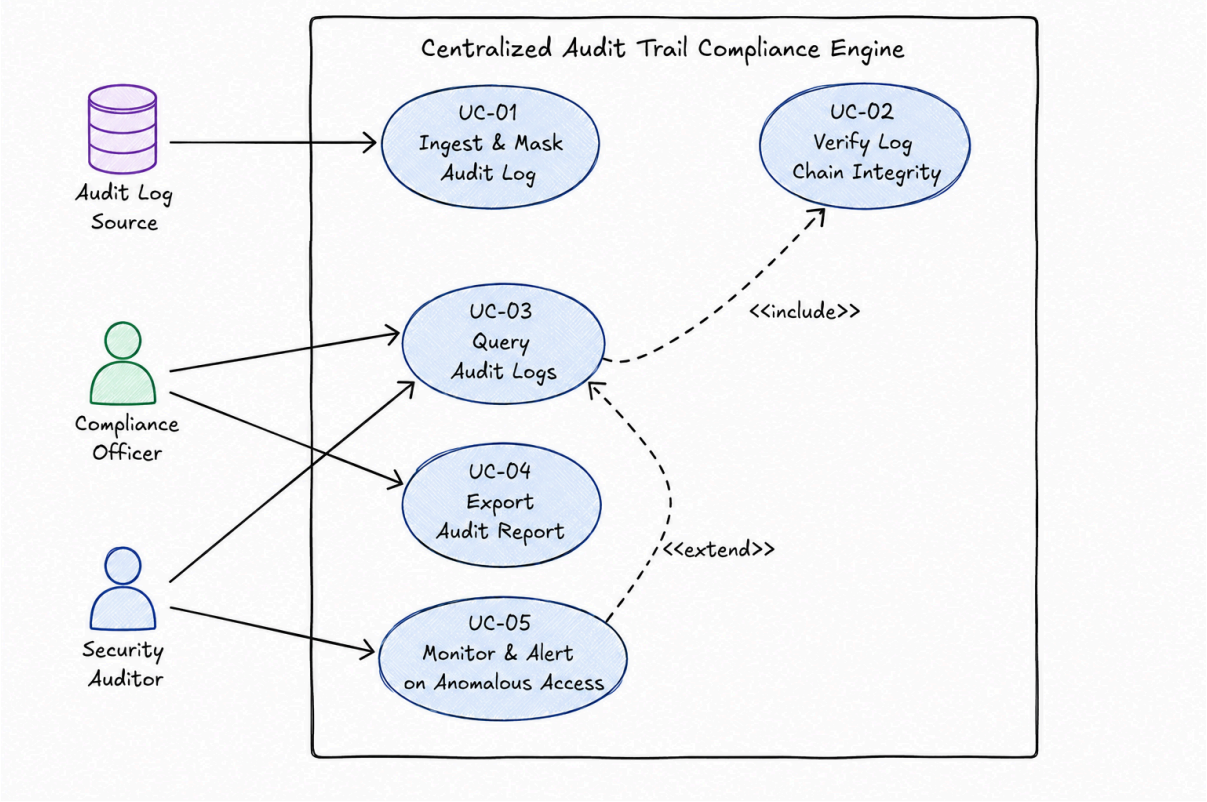
Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001 *(given)*	Functional	The system shall parse structured JSON audit logs, identify and mask PII fields (SSNs, credit cards) before persisting records to storage.	High	Pass: Masked audit log saved with SHA-256 integrity hash; Fail: cleartext credit card stored in DB.	GDPR compliance on ingest.
FR-002	Functional	The system shall generate a cryptographic hash chain (each record's hash includes the previous record's hash) so that any retroactive edit to a stored log is detectable.	High	When any single byte in a persisted record is altered, a chain-verification job flags the record and every subsequent record in the chain as invalid.	Immutability is the core value proposition of an audit trail — without it, logs have no evidentiary weight.
FR-003	Functional	The system shall restrict access to unmasked/raw audit data to users with the "Security Auditor" role; the "Compliance Officer" role shall only see masked views by default.	High	When a Compliance Officer queries a log containing a masked SSN, the field displays as `***-**-1234`; when a Security Auditor queries the same log with elevated access, the full value is shown and the access event is itself logged.	Enforces least-privilege / separation of duties, standard in compliance tooling.

FR-004	Functional	The system shall allow a Compliance Officer to generate an exportable audit report (CSV/PDF) for a specified date range and log source, including the integrity hash for each included record.	Medium	When a report is exported for a given date range, the file contains only records within that range and each row includes its SHA-256 hash.	Exports are an explicit deliverable of the system per the problem context.
FR-005	Functional	The system shall detect and flag anomalous access patterns (e.g., a single user querying unmasked PII fields more than N times in an hour) and raise an alert to the Security Auditor.	Medium	When a user exceeds the configured threshold of unmasked-field queries within 60 minutes, an alert record is created and visible on the Security Auditor's dashboard within 5 seconds.	Compliance systems must monitor misuse of their own privileged access, not just external threats.
NFR-001 *(given)*	Nonfunctional (Performance & Security)	Audit log query endpoints must support complex filtering across 10 million historical records in under 1 second.	High	Pass: benchmark tests confirm target latency and security standards under simulated peak load.	Given.

NFR-002	Nonfunctional (Availability & Durability)	The system shall guarantee zero audit log loss, replicating each ingested record to at least 2 storage nodes before acknowledging ingest as successful.	High	When a single storage node is killed mid-write during a load test, 100% of previously-acknowledged records remain retrievable with valid hashes.	An audit system that can lose records fails its entire regulatory purpose — durability is non-negotiable, distinct from query speed.

UML Use-Case Diagram



Use-Case Flow Document

Actor: Compliance Officer

Includes: UC-02 (Verify Log Chain Integrity)

Preconditions:

- The Compliance Officer is authenticated and holds a valid session with export permissions.
- At least one audit log record exists within the requested date range and log source.

Postconditions (success):

- An export file (CSV/PDF) exists containing only masked fields, each row annotated with its SHA-256 hash.
- The export event itself is written to the audit log (who exported, what range, when).

Main Success Scenario:

1. Compliance Officer selects "Export Report" and specifies a date range and log source.
2. System validates the date range (start ≤ end, within retention window).
3. System retrieves all matching records from storage.
4. System invokes UC-02 (Verify Log Chain Integrity) on the retrieved record set.
5. UC-02 confirms the hash chain is valid for every record in range.
6. System compiles the records into the requested format (CSV/PDF), including each record's masked fields and SHA-256 hash.
7. System writes an audit entry logging the export action (officer ID, range, timestamp).
8. System presents the export file for download.
9. Use case ends successfully.

Alternate Flow

5a. Integrity Verification Fails

- 5a1. UC-02 detects a broken hash chain for one or more records in range.
- 5a2. System halts export generation and displays an "Integrity check failed — export blocked" message, identifying the affected record range.
- 5a3. System automatically raises an alert to the Security Auditor (feeds into UC-05).
- 5a4. Use case ends without producing a file.